Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/19/2019
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	randgerman@gmail.com	**Fax:**	FY none
PI Organization Type:	UNIVERSITY	**Phone:**	858-922-4985
Organization Name:	San Diego State University		
PI Address 1:	College of Engineering		
PI Address 2:	282 Surfview Court		
PI Web Page:			
City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	07/01/2016	**End Date:**	06/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhat, Biliyar	**Contact Phone:**	
Contact Email:	biliyar.n.bhat@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	2019 report: none		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel. NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant #NNX14AB31G with the same Principal Investigator. A research project is in progress to investigate the gravity-dependent densification, microstructure development, and the related macroscopic shape evolution (and distortion) of multi-phase powder materials during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual production of components used in metal cutting, oil well drilling, automotive engines, electric motor bearings, computer heat sinks, jet engine valves, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by sintering. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size.
Task Progress:	[Ed. note 3/15/2019: compiled from PI's annual progress report submitted December 2018] We are investigating the gravity-dependent densification, microstructure development, and macroscopic shape evolution (distortion) during liquid phase sintering. The research focuses on two goals: (i) in-depth analysis of the pore-liquid-grain structure evolution and the role of gravity and (ii) exploring sintering under microgravity conditions as a basis for modeling in-space fabrication or repair. Efforts to extend human exploration to the Moon and beyond require development of processes that permit fabrication under reduced gravity conditions. This will reduce resource requirements for extraterrestrial construction and repair. Many options rely on particulate materials and sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. This research team has developed a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity with tungsten heavy alloys (W-Ni-Cu-Mn) are used to validate the sintering models. The research outcomes are relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon. During the reported period the following research activities have been carried out: * Microgravity sample fabrication and ground testing completed * Flight samples (seven cartridges) were transported to International Space Station on two flights in April and May 2018 * Continuation of modeling of the gravity effects on sintering largely relying on experiments conducted in the ground proof European Space Agency (ESA) Low Gradient Furnace (LGF) * Continuation of fundamental experimentation on the gravity effects in liquid-phase sintering primarily with tungsten heavy alloys with a focus on microstructure gradients in ground samples * Quantitative microstructure analysis versus location in the samples, largely examining density, grain size, and distortion versus tungsten content and dihedral angle * As of 1 Nov 18 samples are awaiting sintering in ESA LGF furnace. The following papers have been submitted for publication: Alvarado-Contreras, J, Maniere, C, German, RM, Olevsky, EA "Multi-scale modeling of various modes of densification of hierarchical powder structures" Submitted Acta Materialia, 2018, submitted and pending review and publication, as of December 2018. Torresani E, German RM, Olevsky EA. "Gravity effects during sintering of additively manufactured powder components." Acta Materialia, 2018, submitted for publication as of December 2018.

Bibliography Type:	Description: (Last Updated: 03/27/2025)
Papers from Meeting Proceedings	German RM, Olevsky EA, Torresani E, Hernandez DL. "Extraterrestrial Sintering." POWDERMET 2018, International Conference on Powder Metallurgy & Particulate Materials, San Antonio, TX, June 17–20, 2018. In: Advances in Powder Metallurgy and Particulate Materials 2018. Princeton, NJ: Metal Powder Industries Federation, 2018. p. 67-80. , Dec-2018